MATH3603 Probability Theory

Test 1

09:00 – 09:50

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1. Suppose A, B, C are independent events. Prove the following four independence results:

(a) A^c and B are independent.

(b) A^c and B^c are independent.

(c) A^c, B^c, and C are mutually independent.

(d) A^c, B^c, and C^c are mutually independent.

2. A stick of length 1 is broken at a point uniformly chosen along its length. Let X be the absolute difference between the lengths of the two resulting pieces, and let Y be the product of the lengths of the two pieces.

(i) Find the distribution of X.

(ii) Find $\mathbb{P}\left(Y < \frac{1}{8}\right)$.

(iii) Find $\mathrm{Var}(X)$ and $\mathrm{Var}(Y)$.

3. There are n people. Each person rolls a fair six-sided die independently.

(i) **Same-value scoring:** If there is a group of identical outcomes, each pair within that group contributes 1 point. For example, if $n = 4$ and the outcomes are $(2, 2, 2, 5)$, then there are $\binom{3}{2} = 3$ pairs of 2's, so the total score is 3. Find the expectation and variance of the total score. Express your answers in terms of n.

(ii) **Multiplicity scoring:** Now the score is defined differently: For each distinct outcome value k (from 1 to 6), if that value appears m_k times, it contributes $m_k \times k$ points. For example, if $n = 4$ and outcomes are $(3, 3, 4, 5)$, the score is $2 \times 3 + 1 \times 4 + 1 \times 5 = 15$; if outcomes are $(2, 2, 2, 5)$, the score is $3 \times 2 + 1 \times 5 = 11$. Find the expectation and variance of this score. Express your answers in terms of n.

End of Test